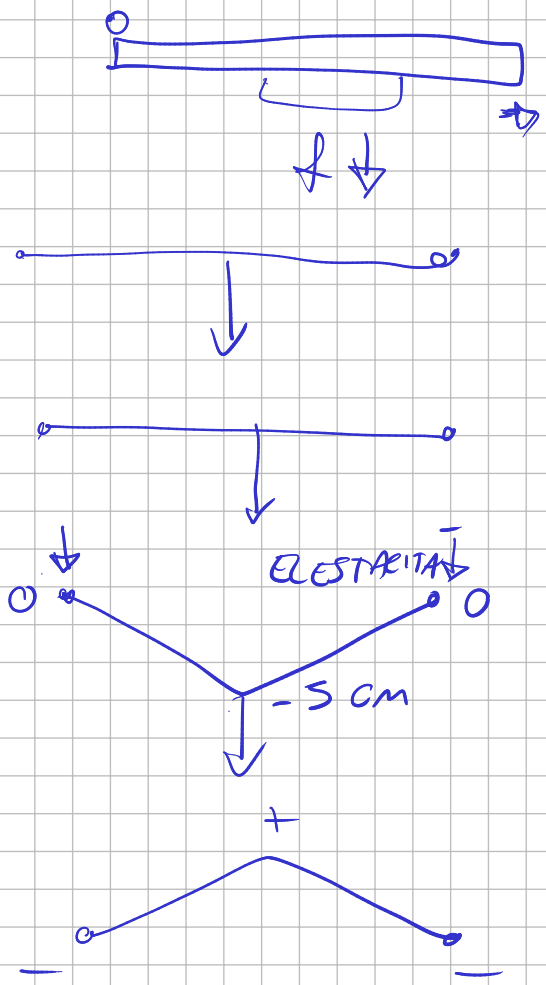
PDE ELLITTICHE (2° ORDINE)



$$-\Delta u = f$$

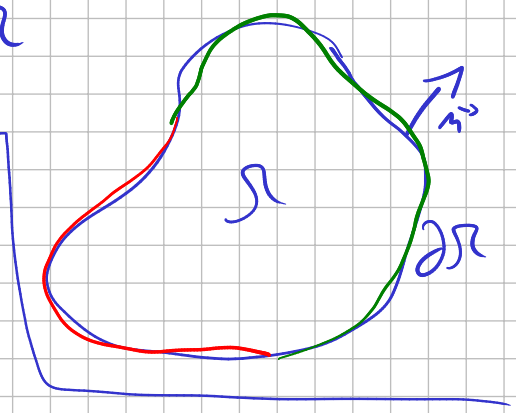
CONDIZIONI DI BORDO

DIRICHLET

$u = g$ su $\partial\Omega$

se $g \equiv 0$ OMOGENEE

NEUMANN $\nabla u \cdot n = \frac{\partial u}{\partial n} = h$ su $\partial\Omega$



$$\partial\Omega = \Gamma_D \cup \Gamma_N$$

$$\begin{cases} -\Delta u = f & \text{su } \Omega \\ u = g_D & \text{su } \Gamma_D \\ \nabla u \cdot n = g_N & \Gamma_N \end{cases}$$

ROBIN $\nabla u \cdot n + \gamma u = r$ su Γ_R

SOLO COND. BORDO NEUMANN

$$\begin{cases} -\Delta u = f & \text{se } u \text{ è soluzione} \\ \nabla u \cdot n = g & u + c \text{ è soluzione} \end{cases} \quad c \in \mathbb{R}$$

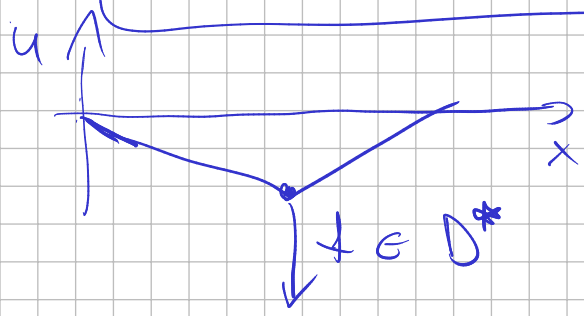
$\Rightarrow$ NON HO UNICITÀ

$$-(\partial_{xx} u + \partial_{yy} u) = f \qquad u \in \mathcal{C}^2(\Omega)\ \mathcal{C}^1(\bar{\Omega})$$

Se $f \in \mathcal{C}^0(\Omega)$, $g_D \in \mathcal{C}^2(\bar{\Omega})$, $g_N \in \mathcal{C}^1(\bar{\Omega})$ $\Rightarrow u \in \mathcal{C}^2$

$$\begin{cases} -u'' = 0 \\ u(0) = 0 \\ u'(0) = 1 \end{cases}$$

$$\boxed{u(x) = x}$$



$$\begin{cases} -u''(x) = \delta_{0.5}(x) \\ u(0) = u(1) = 0 \end{cases}$$

$$u(x) = \begin{cases} -\frac{1}{2}x & x < 0.5 \\ \frac{1}{2}x - \frac{1}{2} & x \geq 0.5 \end{cases}$$

$u'' = 0$ $\qquad$ $u'' = 0$

$$F(x) = \int_0^x f(\xi)\, d\xi \in L^2$$

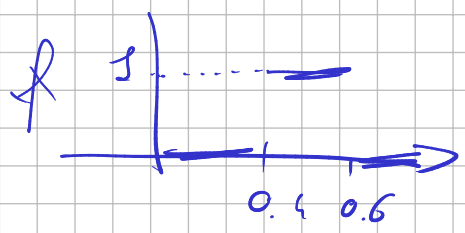
$f \in D^*$

$f \notin L^2$

$f \in L^2$

$$\begin{cases} -u'' = f \\ u(0) = u(1) = 0 \end{cases}$$

$$f(x) = -\chi_{[0.4, 0.6]}(x)$$



$$u(x) = \begin{cases} -\frac{1}{10}x & x < 0.4 \\ \frac{1}{2}x^2 - \frac{1}{2}x + \frac{2}{25} & x \in [0.4, 0.6] \\ -\frac{1}{10}(1-x) & x > 0.6 \end{cases}$$

u è SOLUZIONE

$$-u'' = \begin{cases} 0 & x < 0.4 \\ -1 & 0.4 \leq x \leq 0.6 \\ 0 & x > 0.6 \end{cases} \qquad \begin{aligned} u(0) &= 0 \\ u(1) &= 0 \end{aligned}$$

$$\lim_{x \to (0.4)^+} u(x) = \lim_{x \to (0.4)^-} u(x) \qquad \lim_{x \to (0.4)^+} u'(x) = \lim_{x \to (0.4)^-} u'(x)$$

ANCHE 0.6

$u \in C^1_0$ $\qquad$ $\underline{f \in L^2} \to u \in C^1$ $\qquad$ $\cancel{\in C^2}$

$-u'' = f \qquad \forall\, v \in \underbrace{D(\Omega) = C^\infty_c(\Omega)}_{=0} \qquad \Omega = [0,1]$

$$\int_\Omega \underbrace{-u''\cdot v} = \int_\Omega f\cdot v \Rightarrow \underbrace{-\int_{\partial\Omega} u'\cdot \underbrace{v}_{=0}}_{v \in C^\infty_c(\Omega)} + \int_\Omega u'\cdot v' = \int_\Omega f\cdot v$$

$$\underbrace{\int_\Omega u'\cdot v' = \int_\Omega f\cdot v}$$

$$\begin{cases} u \in H^1_0 \\ v \in H^1_0 \end{cases} \qquad f \overset{\Leftrightarrow}{=} L_f \in (H^1_0)^*$$

$$\int_\Omega f\cdot v = L_f(v) \qquad \forall\, v \in H^1_0$$

$$\boxed{H^1_0 \subset C^0}$$

FORMULAZIONE PDE <u>DEBOLE</u> CON $u \in H^1_0$

$$\int_0^1 u'\cdot v'\,dx = \int_0^1 f\cdot v \qquad \forall\, v \in H^1_0$$

<u>BORDI</u> • DIRICHLET NON OMOGENEO

$$\begin{cases} -u'' = f \\ u(0) = u_L \qquad u(1) = u_R \end{cases} \qquad \Omega = [0,1]$$

$u_{lift}(x) = (1-x)\,u_L + x\,u_R \qquad u''_{lift}(x) = 0$

$u_{lift}(0) = u_L \qquad u_{lift}(1) = u_R$

$\tilde{u} := u - u_{lift} \qquad u = u_{lift} + \tilde{u}$

$$\begin{cases} -\tilde{u}'' = -u'' = f \\ \tilde{u}(0) = 0 \qquad \tilde{u}(1) = 0 \end{cases}$$

POISSON CON BC DIRICHLET OMOGENEE
$\to$ LO CONOSCIAMO GIÀ

$$H^1_0 = \{ u \in L^2 ,\ u' \in L^2 ,\ u(x) = 0 \quad x \in \partial\Omega \}$$

$$\boxed{\int_0^1 u' v' = \delta_{0,5}(v) = v(0.5)} \qquad v \in H^1_0 \subset C^0$$

$\forall v$

NEUMANN NON OMOGENEO + DIRICHLET OMOGENEO

e.s.
$$\begin{cases} -u'' = f \\ u(0) = 0 \\ u'(1) = g_1 \end{cases}$$

$$V = \{ v \in H^1((0,1)) : v(0) = 0 \}$$

$$H^1_0 \subset V \subset H^1 \qquad \forall v \in V$$

$$0 = \int_0^1 -u''\cdot v - f\cdot v \, dx = [-u'\cdot v]_0^1 + \int_\Omega u'\cdot v' \, dx - \int_\Omega f\cdot v \, dx$$

$$= -\underbrace{u'(1)}_{g_1}\cdot \underbrace{v(1)}_{?} + \underbrace{u'(0)\cdot \underbrace{v(0)}_{=0}}_{=0} + \int_\Omega u'\cdot v' \, dx - \int_\Omega f v \, dx$$

$$0 = -g_1 \cdot v(1) + \int_\Omega u'v' \, dx - \int_\Omega f v \, dx$$

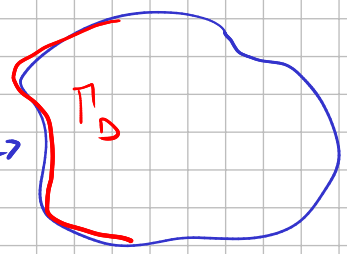
PROBLEMA DEBOLE SU V

TROVA $u \in V$: $\forall v \in V$ VALE

PDE POISSON FORTE IN 2D

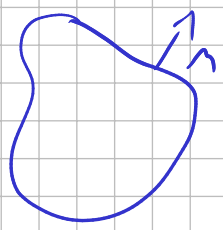
$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u(x) = 0 & \Gamma_D \\ \nabla u(x)\cdot n(x) = g_N(x) & \Gamma_N \end{cases} \qquad \Gamma_D \cup \Gamma_N \subseteq \partial\Omega$$

BORDO DI DIRICHLET →



$$V = \{ v \in H^1(\Omega) : v|_{\Gamma_D} = 0 \}$$

$$\begin{cases} \int_\Omega -\Delta u \cdot v - f \cdot v = 0 \\ BC \end{cases} \qquad \Omega \subset \mathbb{R}^d$$



$$\int_\Omega \operatorname{div}(\vec{a})\, dx = \int_{\partial\Omega} \vec{a} \cdot \vec{n}\, dS$$

$$\parallel \int_\Omega \operatorname{div}(v \cdot \nabla u)\, dx = \int_{\partial\Omega} v \cdot \underbrace{\nabla u \cdot \vec{n}}_{\text{PR SCAMBIO}}\, dS$$

$$= \int_\Omega \sum_{i=1}^d \partial_{x_i}(v\, \partial_{x_i} u)\, dx = \int_\Omega \sum_{i=1}^d \partial_{x_i} v \cdot \partial_{x_i} u + \sum_{i=1}^d v \cdot \partial_{x_i}^2 u\, dx$$

$$= \int_\Omega \nabla v \cdot \nabla u\, dx + \int \Delta u \cdot v\, dx$$

$$-\int_\Omega \Delta u \cdot v\, dx = -\int_{\partial\Omega} v \cdot \underbrace{\nabla u \cdot \vec{n}}_{g_N}\, dS + \int_\Omega \nabla u \cdot \nabla v\, dx$$

FORMULAZ. DEBOLE È TROVA $u \in V \subset H^1$: $\forall v \in V$

$$\int_\Omega \nabla u \cdot \nabla v\, dx = \int_\Omega f \cdot v\, dx + \int_{\Gamma_N} v \cdot g_N\, dS$$

$$\int_{\partial\Omega} v \cdot \nabla u \cdot \vec{n}\, dS = \int_{\Gamma_D} \underbrace{v}_{=0} \cdot \nabla u \cdot \vec{n}\, dS + \int_{\Gamma_N} v\, \underbrace{\nabla u \cdot \vec{n}}_{g_N}\, dS$$

POISSON + μ COEFFICIENTE (CONDUCIBILITÀ TERMICA O SIMILI)

+ σ COEFF. DI REAZIONE

Ω $\qquad \partial\Omega = \Gamma_N \cup \Gamma_D \qquad \Gamma_N \cap \Gamma_D = 0$

$$\begin{cases} -\operatorname{div}(\mu \nabla u) + \sigma u = f & \text{in } \Omega \\ u = u_D & \Gamma_D \\ \nabla u \cdot n = g & \Gamma_N \end{cases}$$

$$u_{lift} \in V_D = H^1(\Omega) \cap \{ v : v|_{\Gamma_D} = u_D \}$$

$$\tilde{u} \in V = \underline{H^1_0(\Omega)}$$

$$u = u_{lift} + \tilde{u}$$

$$\int_\Omega \mu \nabla u \nabla v - \int_{\partial\Omega} v \cdot \mu \nabla u \cdot n \, dS + \int_\Omega \sigma u \cdot v = \int_\Omega f \cdot v$$

$$\int_\Omega \mu \nabla u_{lift} \cdot \nabla v + \int \mu \nabla \tilde{u} \cdot \nabla v + \int_\Omega \sigma \tilde{u} \cdot v + \int_\Omega \sigma u_{lift} v$$

$$= \int_\Omega f \cdot v + \underbrace{\int_{\Gamma_D} v \underbrace{\mu \nabla u_{lift} \cdot n}_{=0}}_{=0} + \int_{\Gamma_D} \underbrace{v}_{0} \mu \nabla \tilde{u} \cdot n$$

$$+ \underbrace{\int_{\Gamma_N} v \cdot \mu \nabla u_{lift}}_{(1)} + \underbrace{\int_{\Gamma_N} v \cdot \mu \nabla \tilde{u}}_{(2)}$$

CASO 1 $\quad \mu \nabla u_{lift} \cdot n = 0 \Rightarrow (1) = 0$

$$\boxed{\text{CASO 2} \quad \mu \nabla \tilde{u} \cdot n = 0 \Rightarrow (2) = 0}$$

PIÙ STANDARD

$$a(u,v) := \int_\Omega \mu \nabla u \nabla v \, dx + \int_\Omega \sigma u \cdot v \, dx$$

$$F(v) := \int_\Omega f v \, dx + \int_{\Gamma_N} g v \, dS - \int_\Omega \mu \nabla u_{lift} \cdot \nabla v \, dx - \int_\Omega \sigma u_{lift} v$$

a BILINEARE $\quad$ F LINEARE

F LINEARE ✓ $\quad$ LIMITATO ✓

a BILINEARE ✓ $\quad$ SIMMETRICA ✓

a CONTINUA $\quad |a(u,v)| \leq C \|u\|_V \cdot \|v\|_V$

a COERCITIVA $\quad |a(u,u)| \geq \alpha \|u\|_V^2$

TEOREMA LAX-MILGRAM

SIA V SP. HILBERT, $a(\cdot,\cdot) : V \times V \to \mathbb{R}$ BILINEARE CONTINUA e COERCITIVA, SIA $F : V \to \mathbb{R}$ LINEARE LIMITATA $\Rightarrow \exists!$ SOLUZIONE AL PROBLEMA:

(*) TROVA $u \in V$: $\underline{a(u,v) = F(v)}$ $\forall v \in V$.

COROLLARIO

LA SOLUZIONE u di (*) È LIMITATA DA $\|u\|_V \leq \frac{1}{\alpha} \|F\|_{V^*}$

DIM

$$\alpha \|u\|_V^2 \leq a(u,u) \underset{\text{scelgo } v=u}{=} F(u) \leq \|F\|_{V^*} \cdot \|u\|_V$$

$$\alpha \|u\|_V \leq \|F\|_{V^*}$$

$\boxtimes$

(IL NOSTRO PROBLEMA È BEN POSTO)